

Software Engineering — Assignment 2

Agile Project Management using Scrum in Jira

TEAM NAME

Team Runtime Terror

PROBLEM STATEMENT

Civic/Community Issue Reporting & Routing System

TEAM MEMBERS

Anish K V CB.EN.U4CCE24005
anlietyt@gmail.com

Balaji M CB.EN.U4CCE24010
balajim2209@gmail.com

Deliverables in this submission

Jira site: <https://anlietyt.atlassian.net> · Project key: CIRRS · Board: CIRRS board (Scrum, company-managed)

Deliverable	Where it is	Type	Status
1. Jira Project Screenshots / Export	Section 1 of this document	Screenshots	Included
2. Product Backlog	Excel workbook (submitted separately)	Spreadsheet	Included
3. Sprint Backlog	Excel workbook (submitted separately)	Spreadsheet	Included
4. Scrum Board Screenshots	Section 2 of this document	Screenshots	Included
5. Burndown Chart Screenshot	Section 3 of this document	Screenshot	Included
6. Sprint Retrospective Report	Section 4 of this document	Report	Included

1. Jira Project Overview

The project summary view shows the overall state of the CIRRS project: 121 total work items with their status distribution, the priority breakdown across the backlog, the mix of work item types, and recent activity on the board.



Figure 1 — Jira project summary: status overview, priority breakdown, work item types and recent activity

2. Scrum Board

The Scrum board tracks work through the workflow stages To Do → In Progress → In Review → Done. Work items are grouped by user story, so each story sits in the column matching its own status with its sub-tasks listed beneath it.

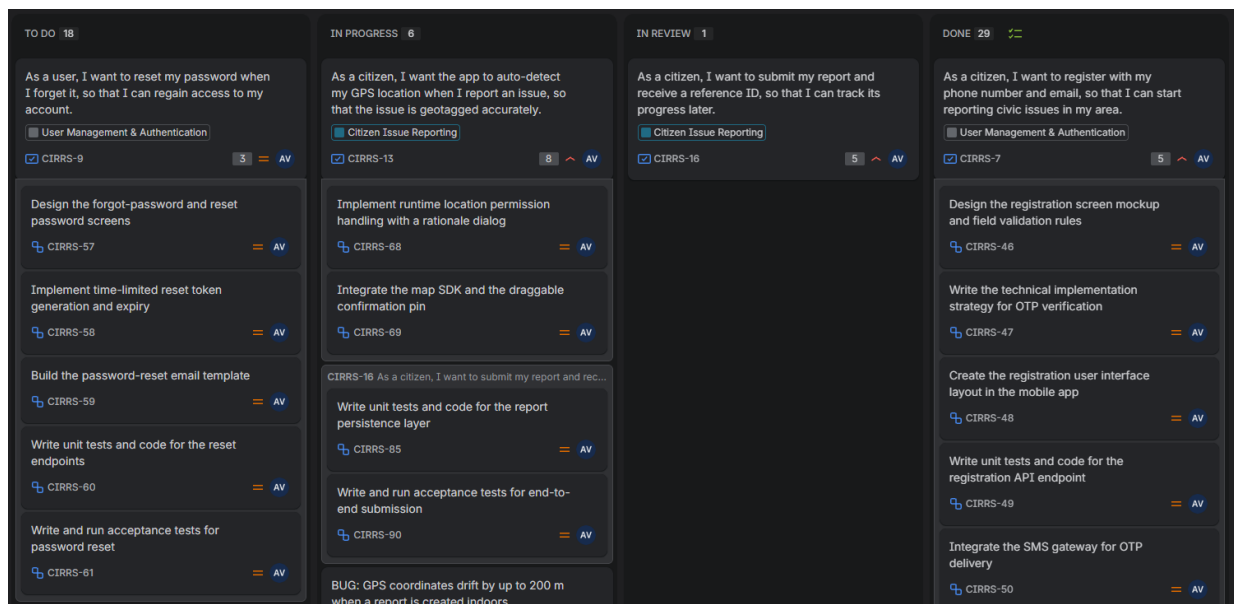


Figure 2 — CIRRS Sprint 1 Scrum board: 18 To Do, 6 In Progress, 1 In Review, 29 Done

3. Sprint Burndown Chart

The burndown chart for CIRRS Sprint 1 (24 August – 7 September 2026) tracks remaining story points against the sprint guideline. The sprint was committed at 34 story points.

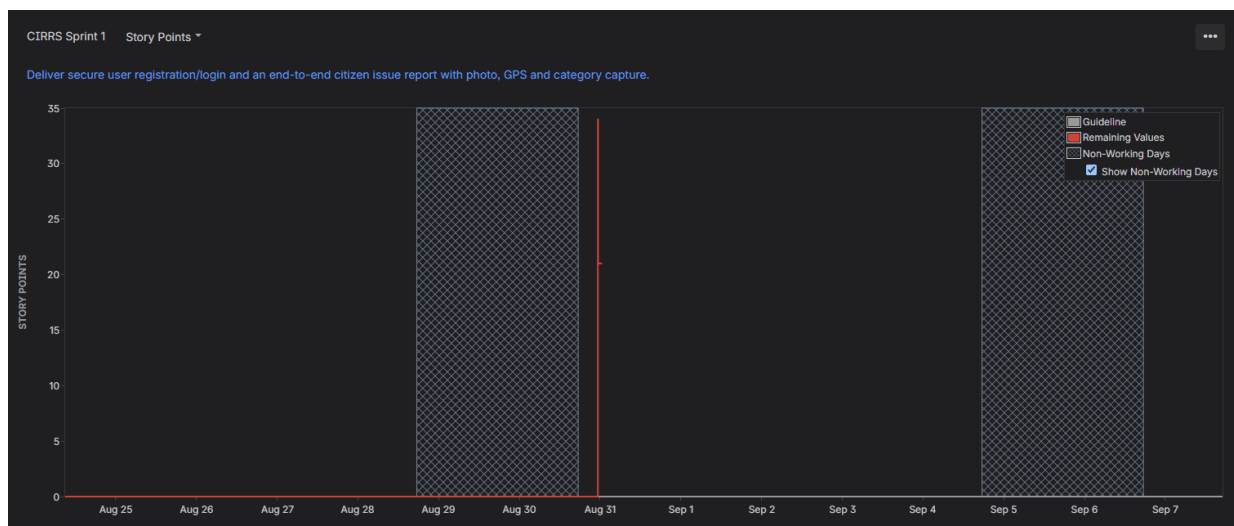


Figure 3 — CIRRS Sprint 1 burndown chart, measured in story points

4. Sprint Retrospective Report

CIRRS Sprint 1 · 24 August – 7 September 2026 · Team Runtime Terror

4.1 Sprint overview

Sprint	Dates	Committed	Completed
CIRRS Sprint 1	24 Aug – 07 Sep 2026	34 story points	13 story points
Work items	54 (incl. sub-tasks)	Done: 29	In progress: 6
Jira project	CIRRS (Scrum board)	Epics: 6	Stories: 29

Sprint goal: deliver secure user registration and login, and an end-to-end citizen issue report with photo, GPS and category capture.

This retrospective was held at the sprint mid-point review. Registration (CIRRS-7), login (CIRRS-8) and photo capture (CIRRS-12) are Done; report submission (CIRRS-16) is in review and GPS auto-detection (CIRRS-13) is in progress.

4.2 What went well during the sprint

- Breaking every user story into five or six sub-tasks made progress visible daily. Registration and login moved through To Do → In Progress → Done without either of us having to ask what the other was working on.
- Splitting ownership by epic worked cleanly. Anish took authentication, citizen reporting and notifications; Balaji took municipal triage, department resolution and the admin dashboard. There were no merge conflicts because the two areas touch different components.
- Writing acceptance criteria into each story before development started meant "done" was never argued about. The five criteria on CIRRS-7 were what we actually tested against.
- Labelling issues by actor (Citizen, Staff, Officer, Admin, User-Admin) and tagging components (Mobile-App, Web-Portal, Backend-API, Database, Notification-Service) made the backlog filterable, so sprint planning took under an hour.
- The three foundation tasks — repository and CI setup, database schema, and wireframes — were finished in the first week, which unblocked all the feature work behind them.

4.3 Challenges encountered

- GPS location capture (CIRRS-13, 8 points) was under-estimated. Runtime permission handling, reverse geocoding and the draggable map pin each turned out to be separate pieces of work, and the story is still open at the mid-point.
- Two defects surfaced during testing rather than development: GPS coordinates drift by up to 200 m indoors (CIRRS-42) and photo uploads fail silently above 5 MB (CIRRS-43). Both were logged as bugs and pulled into the sprint, which pushed the password reset story (CIRRS-9) to the back of the queue.
- Password reset (CIRRS-9) has not been started at all. It depends on an email service that was not arranged during sprint planning, which is a planning gap rather than a development problem.
- Story point estimates were guesses in the first sprint because we had no velocity history to calibrate against. The 8-point stories in particular were closer to 13.

4.4 Areas for improvement

- Estimate against a reference story. Now that CIRRS-7 is a known 5 points, future estimates should be sized relative to it instead of from scratch.

- Identify external dependencies (SMS gateway, email service, map provider quota) during sprint planning and raise them as tasks, so no story enters a sprint blocked.
- Test on real devices earlier. Both Sprint 1 defects would have been caught in the first week rather than the second if acceptance testing had run alongside development instead of after it.
- Cap the sprint commitment at measured velocity. Sprint 1 committed 34 points; Sprint 2 currently holds 48, which should be groomed down once actual Sprint 1 velocity is known.
- Carry unfinished work back to the backlog for re-prioritisation rather than rolling it forward automatically.

4.5 How Scrum practices helped planning, collaboration and execution

Planning. Writing the work as six epics mapped directly onto the modules identified in Assignment 1 — the use case, activity and class diagrams translated almost one-to-one into epics and stories. Allocating those stories across three sprints, each aligned to a release version, turned an abstract design document into a schedule with dates.

Collaboration. The Scrum board gave both of us a shared picture of the work without a status meeting. Because every item is assigned and every story has acceptance criteria, handovers were unambiguous. Assigning by epic rather than by task also gave each of us end-to-end ownership of a feature area, which reduced the amount of coordination needed.

Execution. The To Do → In Progress → In Review → Done workflow made blocked work obvious: an item sitting in In Progress for several days is visible to both of us on the board. The burndown chart showed the gap between planned and actual progress mid-sprint, which is what prompted the scope conversation captured in this retrospective rather than discovering the shortfall on the last day.

4.6 Actions for Sprint 2

- Split CIRRS-13 (GPS auto-detection) into two stories — location capture, and map confirmation — before it carries over.
- Add an external-dependency checklist to the sprint planning agenda; raise the email service for CIRRS-9 as a task now.
- Re-estimate the Sprint 2 backlog against CIRRS-7 as the 5-point reference and groom it down towards measured velocity.
- Move acceptance testing sub-tasks to run in parallel with development rather than at the end of each story.
- Review the two open backlog defects (CIRRS-44, CIRRS-45) at the next grooming session and schedule them.